



PHYSICAL AI: WHOLE-CURRICULUM ARCHITECTURE ROADMAP

A Principled Engineering Journey from Physical Constraints to Frontier Dual-Brain Systems

⌘ PART I · FOUNDATIONS

Physics, Silicon & Hierarchy

Ch 1: The Boundary

The Physical AI Imperative · Causal Loop

Ch 2: Constraints

5 Physical Columns · Latency Freshness Wall

Ch 3: Cognition

5 Cognitive Stages · The Great Systems Tension

Ch 4: Hierarchy

Multi-Rate Topology · Three Cadences

Formulate

🔧 PART II · THE 5 ORGANS

Sensory-Motor Execution Pipeline

Ch 5: Perception

Sensory Transduction · Spatial Tokens

Ch 6: Memory & State

SE(3) World Models · Frame Trees

Ch 7: Reasoning & Intent

VLM Grounding · Expiring Intent Leases

Ch 8: Planning & Synthesis

Diffusion / ACT · C^2 Jerk Continuity

Ch 9: Reflex Enforcement

1 kHz MCU Enforcer · CBF Safe Sets

Deploy

🛡️ PART III · RELEASE

Hardware, Governance & Release

Ch 10: Placement

4-Tier Placement · Thermal Budgets

Ch 11: Governance

Authority Arbitration · Bumpless Override

Ch 12: Assurance

CAE Safety Cases · 4-Rung Qualification

Ch 13: The Frontier

Dual-Brain Kit · Oral Defense Dossier